Mapping Refugee Settlement Growth and Resource Stress in Lebanon's Bekaa Valley: A Geospatial Analysis of Displacement Impact

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1. Introduction

Since the onset of the Syrian civil war in 2011, Lebanon's Bekaa Valley has become a major host region for displaced populations. According to UNHCR data, Bekaa hosts approximately 265,892 registered refugees, accounting for 37.1 percent of the 772,000 officially registered refugees in Lebanon, although the Government of Lebanon estimates the total refugee population to exceed 1.4 million (UNHCR, 2025). An estimated 1,600 informal settlements have been established across the valley to accommodate this population (Medair, 2023).

This study analyzed the spatial expansion of these settlements over the past decade and assessed their environmental impact using geospatial tools. Specifically, it examined changes in land cover and vegetation health surrounding refugee camps between 2013 and 2024, focusing on farmland conversion and indicators of vegetation and moisture stress derived from NDVI and NDMI.

The rationale for this work lies in the urgent need to understand how displacement-driven land use affects fragile ecosystems and resource availability. The Bekaa Valley is agriculturally vital and environmentally sensitive. The rapid influx of refugees has placed sustained pressure on water resources, waste management systems, and local biodiversity. While humanitarian mapping efforts have largely focused on infrastructure and logistics, few studies have quantified the long-term ecological consequences of settlement growth using satellite data.

The findings have potential to contribute spatial evidence for understanding settlement-driven land use change. The findings could provide insight for humanitarian agencies, environmental planners, and policymakers seeking to balance refugee support with sustainable land management in the Bekaa Valley.

2. Research Objectives/Questions

This report meets the following objectives:

1. To map the spatial growth of 5 informal refugee settlements within one irrigation control zone in the Bekka valley from 2013 to 2024.

2. To assess land cover change/land use change within and around 200 meters of each camp.
3. To assess water stress on irrigated agriculture in the same irrigation control zone as the camps.
4. To look for correlations between informal settlement expansion and ecological stress indicators.

3. Methods/Materials

3.1 Study Area/Parameters

For this study, I looked at informal settlements in the Bekka valley. I decided to focus my spatial analysis on 6 settlements which had at least 150 tents and were within the same irrigation control zone. To delineate my study area, I filtered the camps layer in ArcGIS Pro to display only settlements with 150 tents or more, then exported that filtered layer to Google Earth Pro (GEP) to take advantage of its historical imagery and high-resolution basemaps. Using GEP, I identified six large camps located close together. I focused my vegetation health and water stress analysis on the irrigated extent of the Upper Litani Basin, which will be discussed further in section 3.4.

3.2 Data

Syrian refugee population data for registered refugees as well as estimates of real refugee population were obtained from UNHCR (2025). BucknellGIS (2023) hosts a dataset which contains information on informal refugee settlements in the Bekka valley, that I downloaded from ArcGIS living atlas.

For imagery, I used Landsat-8 Collection 2 for all my 2013-2014 images. I used Sentinel-2 Level 1-C for all images 2015 and later. These image collections were accessed, processed, or exported via Google Earth Engine (GEE). I ensured that the code follows standards which I have learned in previous courses, such as applying atmospheric correction, using correct parameters to filter composite images by date, etc. Land cover data for 2013 were obtained from the European Space Agency (ESA) Climate Change Initiative (CCI) Land Cover dataset, while 2024 land cover data were sourced from the Copernicus Global Land Service (CGLS) Land Cover product. Both were accessed and exported via GEE.

3.3 Land Use Analysis

I used GEP to measure the spatial growth of settlements from the start of the Syrian conflict in 2013 to 2024. I used the historic imagery slide tool to move between dates in the same scene. I drew polygons of the camp boundaries for every year, and measured the area in hectares of the polygon directly on GEP (figure 3.3a) Then, I used to GEE to access 2015 and 2024 imagery, and compare them side by side using a swipe widget (figure 3.3b) (view in GEE here).

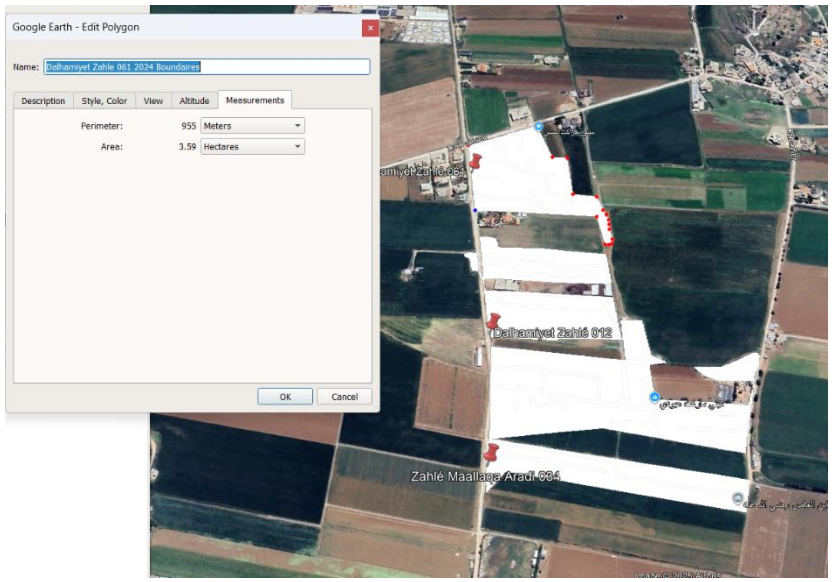

Figure 3.3a. Camp Settlement GEP method

Figure 3.3b. 2015 to 2024 comparison slider.

The land use change quantification was done in ArcGIS Pro by first overlaying my 2024 camp polygons on my 2013 land use raster. I buffered my polygons 200 meters using a geodesic preserving buffer, and included the area within the polygon in my analysis. Using the dissolve tool in ArcGIS, I created one buffer area for 3 adjacent camps, to ensure that overlapping buffer areas do not get counted twice. After checking my 2013 raster and ground proofing with GEP high-resolution imagery, I saw that every polygon was farmland in 2013. (figure 3.3c), I only needed to quantify the built-up area in my 2024 raster. I ran tabulate area to find the number of meters classified as built-up in 2024, and divided that by 10,000 to see how many hectares that translates to (figure 3.3d)

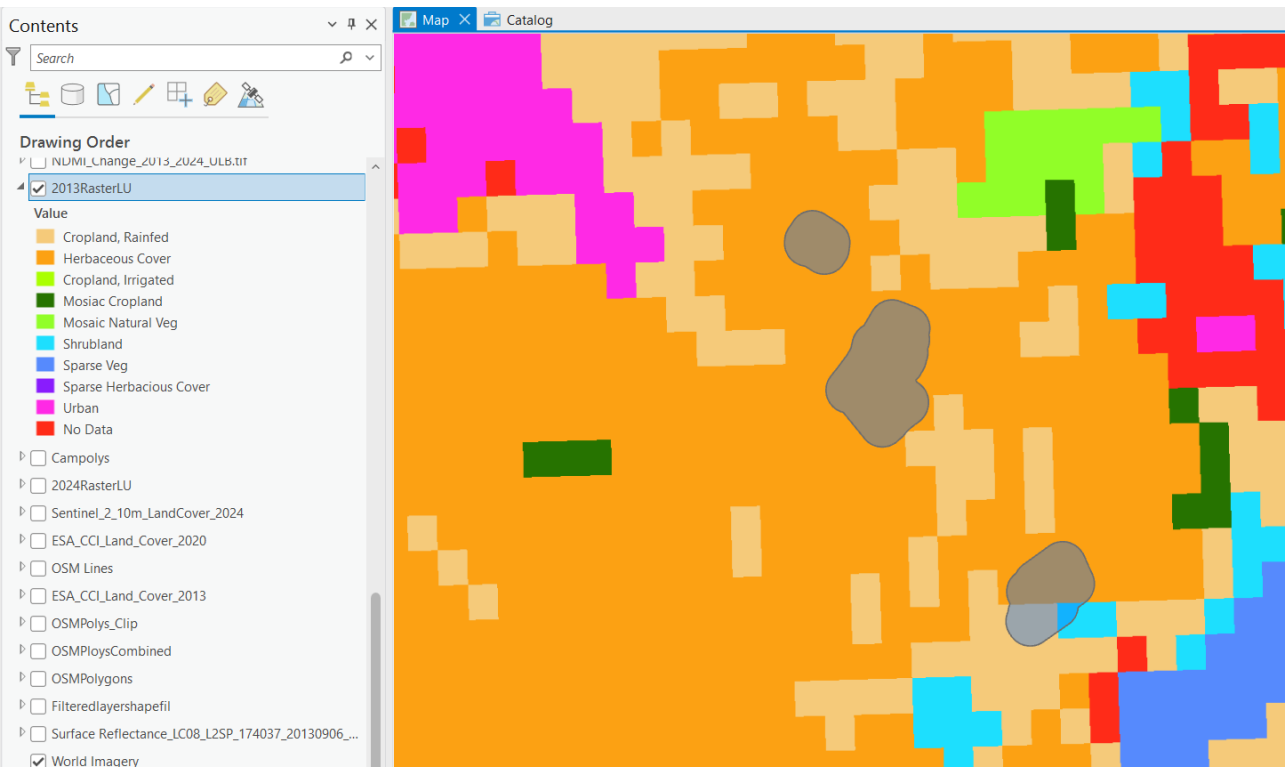

Figure 3.3c. 2013 Land use raster with camp buffer area.

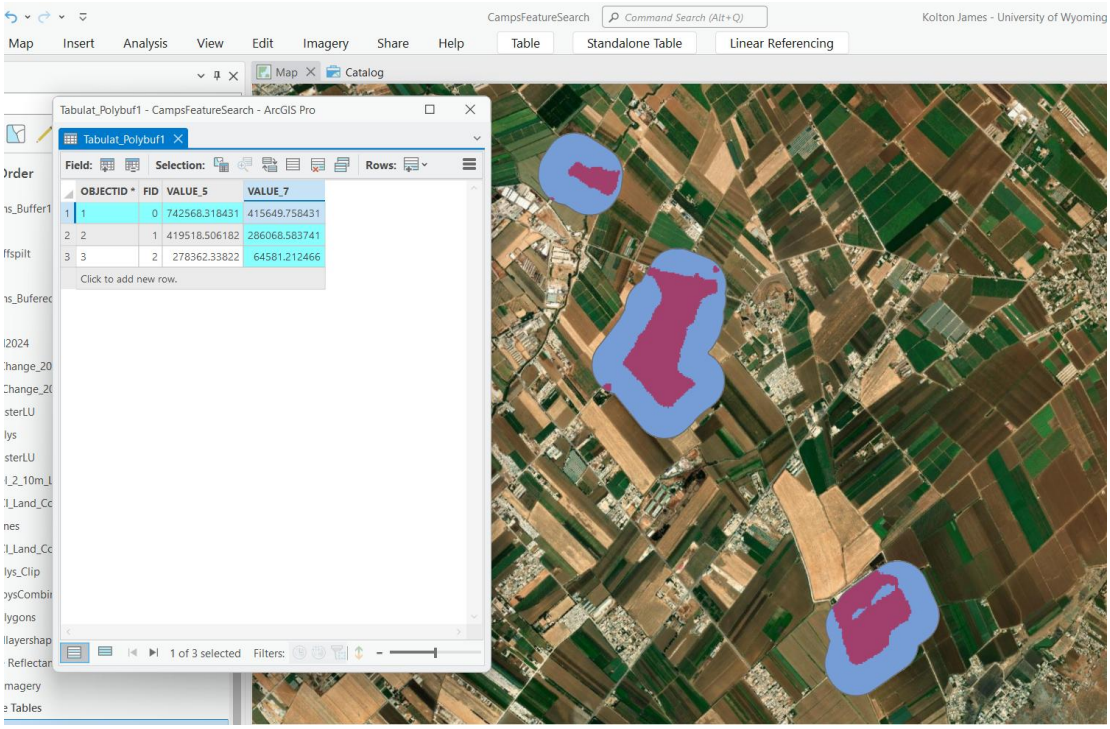

Figure 3.3d 2024 Land use raster clipped to camp buffer area.

3.4. Vegetation and Water Stress Analysis.

The agricultural impacts of increased domestic water consumption will extend across the irrigation control zone. The irrigation control zone (or irrigation command area) is the extent of irrigated area that shares the same system of water resources (United Nations 2025).

I delineated the irrigation control zone based on the study by Jaafar et al. (2016), which treats the Upper Litani Basin as a unified hydrological and management system encompassing surface water, groundwater and irrigation agriculture. I used GEE to create a summer NDVI map, and drew a polygon around the irrigated area(feature 3.4a) (view in GEE here). Irrigated fields appear yellow and green compared to non-irrigated land, which appears red.

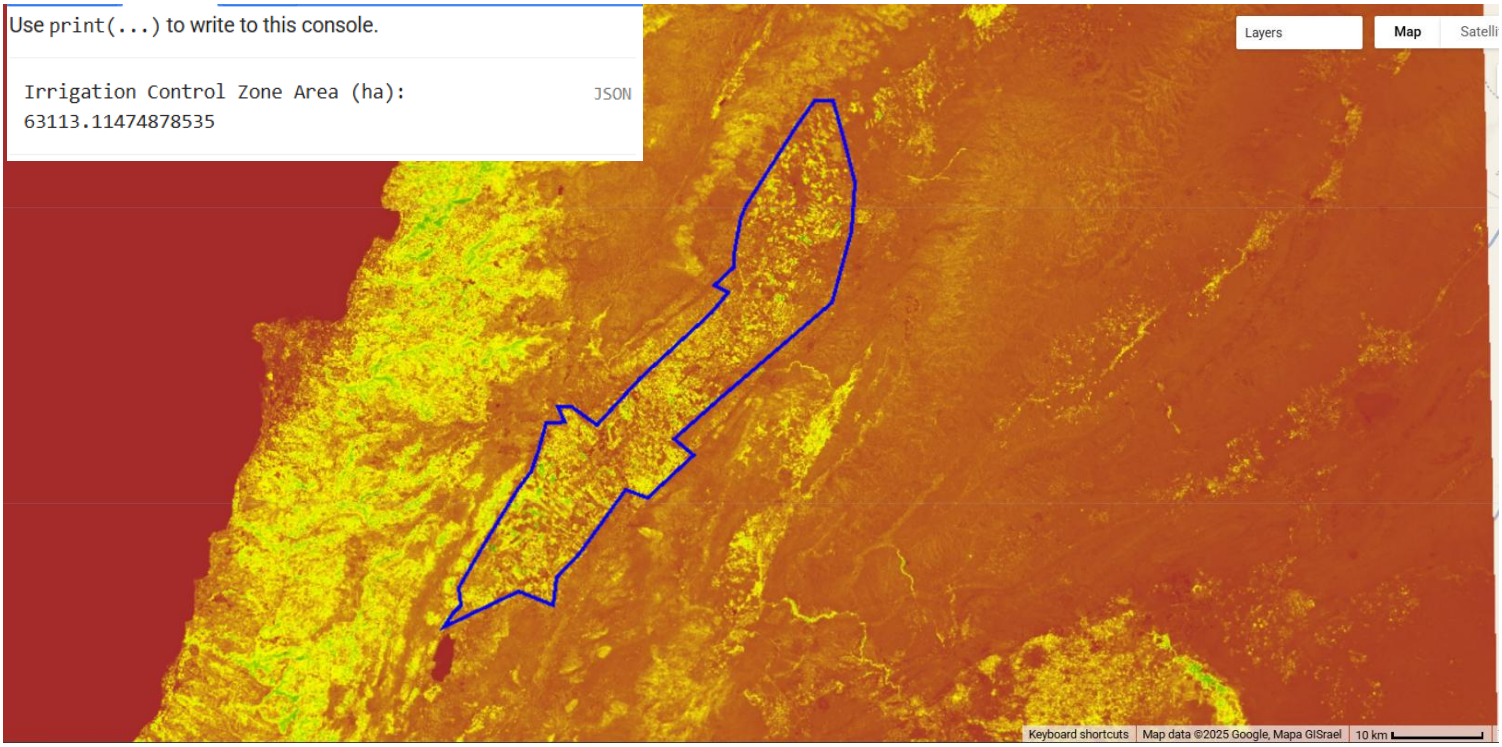

Figure 3.4a. 2013 NDVI map with irrigation control zone polygon.

The Normalized Difference Moisture Index (NDMI) measures plant moisture content and can be used as an indicator of water stress. (Safi et al., 2022) The Normalized Difference Vegetation Index (NDVI) measures plant greenness and can be used as an indicator for plant health (USGS). NDVI and NDMI, when combined together, can give better insight into the condition of crops than just NDVI or NDMI alone. (Varouchakis et al., 2025)

To find the mean NDVI/NDMI values for each year, I built annual summer (June–August) median composites in Google Earth Engine using cloud-masked Landsat 8 and Sentinel-2 imagery. Each composite was clipped to the Upper Litani Basin irrigation control zone and included NDVI and NDMI bands derived from the near-infrared and red/SWIR bands. Mean NDVI and NDMI values were then extracted for each year by applying reduceRegion(ee.Reducer.mean()) over the irrigation control zone.

I created 2013–2024 change maps (figure 4.3e and 4.f) by subtracting the 2013 NDVI and NDMI composites from the 2024 composites. The 2024 composites were reprojected to 30m resolution in order to compare with the 2013 composites (view in GEE here).

3.5 Correlation/Relationship Analysis

To examine how refugee settlement expansion may relate to ecological stress, I compared changes in camp area with NDVI and NDMI trends from 2013 to 2024. The mean NDVI and NDMI values extracted from GEE were plotted as line charts (figure 4.3a and 4.3b) to show vegetation and moisture trends over time within the Upper Litani Basin irrigation control zone. Downward trends in NDVI and NDMI are interpreted as indicators of declining vegetation health and increasing water stress across irrigated farmland. I created an aggregate dataset with mean values per year and total camp hectares per year. This dataset was used to make scatterplots of mean NDVI and NDMI with total camp hectares (figure 4.3c and 4.3d). All charts and graphs were created by using exported and manually created excel sheets. After generating the visualization, I performed visual analysis to look for correlations and trends.

4. Results and Discussion

4.1 Settlement Growth

All six camps in the study area were established by 2014, with several appearing as early as 2013. The largest expansion periods occurred between 2014–2015 and 2016–2017. Camps 034, 012, 155, and 061 showed the most rapid growth during 2014–2015, while the 028/032 cluster experienced its main expansion between 2016–2017. By 2018, all camps had reached their maximum spatial extent, totaling approximately 22.63 hectares. Which is an average of 3.77 hectares per camp (counting 028 and 032 separately). From 2018 to 2024, the total footprint of the camps remained stable.

The timing of settlement growth is likely due to escalations of the Syrian civil war, or its expansion into previously undistributed areas, which caused local populations to flee.

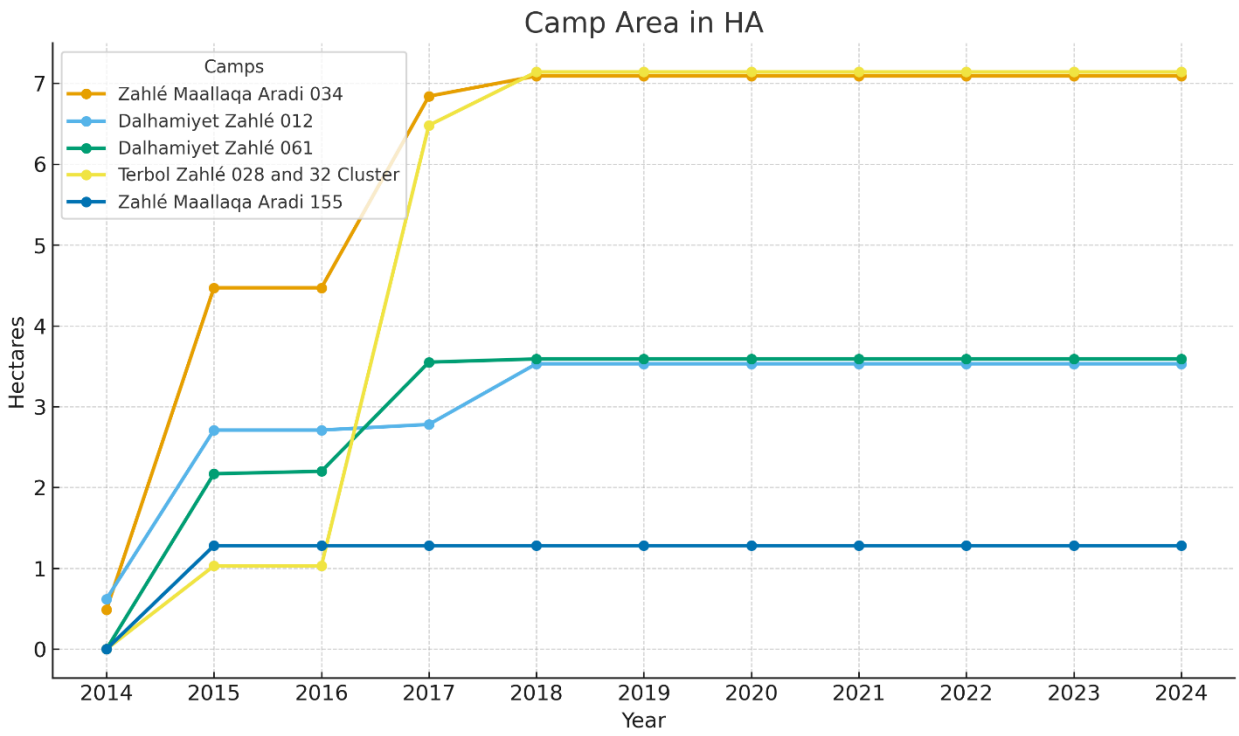

Figure 4.1. Settlement growth by year, 2013–2024

4.2 Land Use Change

All 6 camps that I studied were established on agricultural land. On average, 12.78 hectares of farmland was converted within 200 meters of each camp (figure 4.2). The (BucknellGIS 2023) dataset contains 143 camps that include between 50 and 244 tents. If we were to roughly estimate based on the average for my 6-camp sample, there could be as much as 1,827 hectares of farmland converted in the Bekka valley due to the refugee crisis. As this isn't a complete dataset, the total farmland loss could be even higher.

My analysis has some limitations. I did not perform an accuracy assessment of my land use raster, and so some pixels could have been misrepresented. Also, it is difficult to determine the association of build-up areas around the camp to the camp itself. Medair (2023) mentions how schools and other infrastructure are sometimes placed around the camp. The most common methods of water discharge are directly into the environment. While many camps have municipal trash collection, many camps burn garbage (BucknellGIS 2023). Infrastructure such as schools, solar arrays, roads, well as water and waste disposal strategies may explain farmland loss adjacent to camps.

The camp could have contributed to the new build-up within its buffer, but other factors could be involved. Such as factors relating to normal development or urbanization of this region.

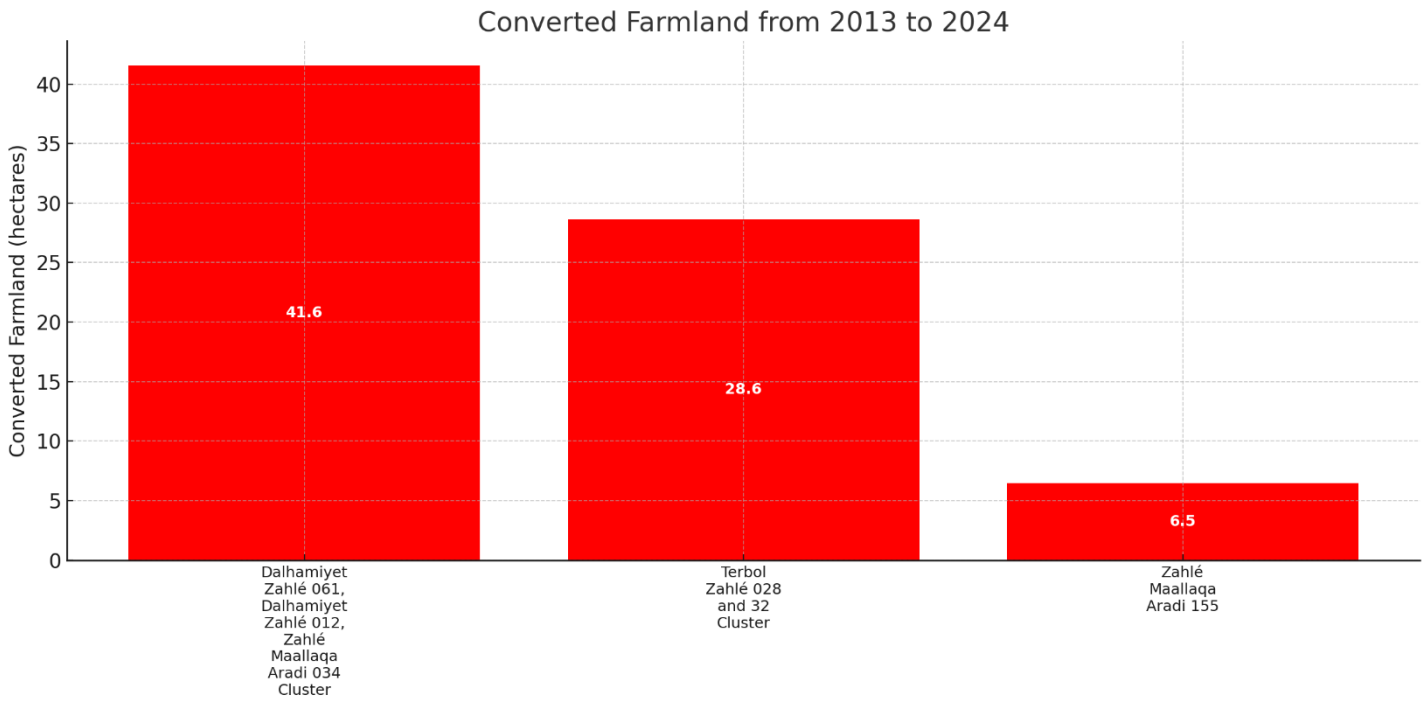

Figure 4.2 Converted Farmland from 2013 to 2014

4.3 Vegetation Health and Water Stress Indicators

Mean Yearly Values

Looking at the line graphs of yearly mean values (figure 4.3a and 4.3b), we can see some clear trends. From 2013 to 2019, NDMI experienced greater fluctuation than NDVI. Before 2019, both NDVI and NDMI had ups and downs with no sustained trend. Both NDVI and NDMI experienced a sharp rise from 2018 to 2019. After 2019, both NDMI and NDVI began a dramatic and sustained downward trend, with NDVI falling to its decade low by far, and NDMI falling to nearly its decade low.

The relatively stable NDVI and NDMI values between 2013 and 2018, followed by a dramatic and sustained decline after 2019, could be related to the expansion of informal settlements and the resulting pressure on agricultural and water resources.

Although official refugee registration in Lebanon was frozen in 2015, field data as well as my own spatial analysis indicate that informal settlement growth in the Bekaa Valley continued through 2018, reaching a peak in density and land area footprint during that period (UNHCR, 2018; Medair, 2023; King-Okumu et al., 2016).

In Jaafar, Archer, et al (2016), the author notes that "In some municipalities, the population and its demand for water have tripled in size since the arrival of displaced people." After 2019, the cumulative domestic water demand from refugees and host communities could have begun to exceed the capacity of the irrigation network. This would lead to increasing groundwater withdrawals and reduced water availability for agriculture, conditions which are consistent with the negative NDVI/NDMI trend observed in the following years.

The sharp rise from 2018 to 2019 is most likely linked to higher-than-average rainfall that year. According to ERA5 monthly precipitation anomaly data compiled by Our World in Data (2024), Lebanon experienced wetter-than-average conditions during the winter of 2018–2019. UNHCR (2019) reported heavy rainfall and widespread flooding from Storm *Norma* in January 2019, consistent with the NDVI/NDMI spike observed for that year.

Change Detection Maps

The NDMI change map (figure 4.3f) has a lot more areas of no change than the NDVI change map (figure 4.3e). NDMI appears to have less of a dominance of red over green, while NDVI has is much more predominately red than green. NDVI has almost no farmland that didn't change, while NDMI had farmland that experienced no change.

One thing to consider about these maps is that while there are a lot of green fields present, the green that is present is less intense in color than the red that is present. It looks at first glance that there are only slightly more red areas than green. However, the red areas are usually intensely red, indicating a more dramatic negative change. The green areas are usually only very light to light green, indicating only a small positive change.

In terms of spatial distribution, the greenest areas in both maps seem to be the southernmost point of the Upper Litani Basin. The northernmost areas have the second most proportion of green for both maps. Observed differences in spatial distribution of positive and negative change could reflect local agricultural management strategies, such as crop rotation or periodic field fallowing.

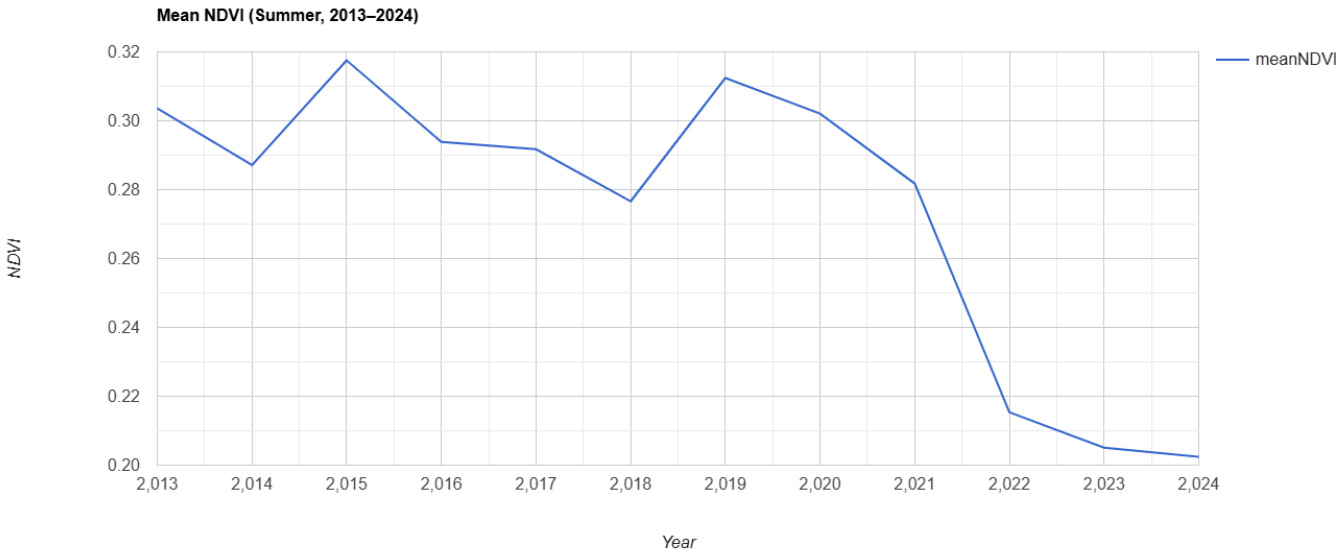

Figure 4.3a Mean Summer NDVI 2013-2024

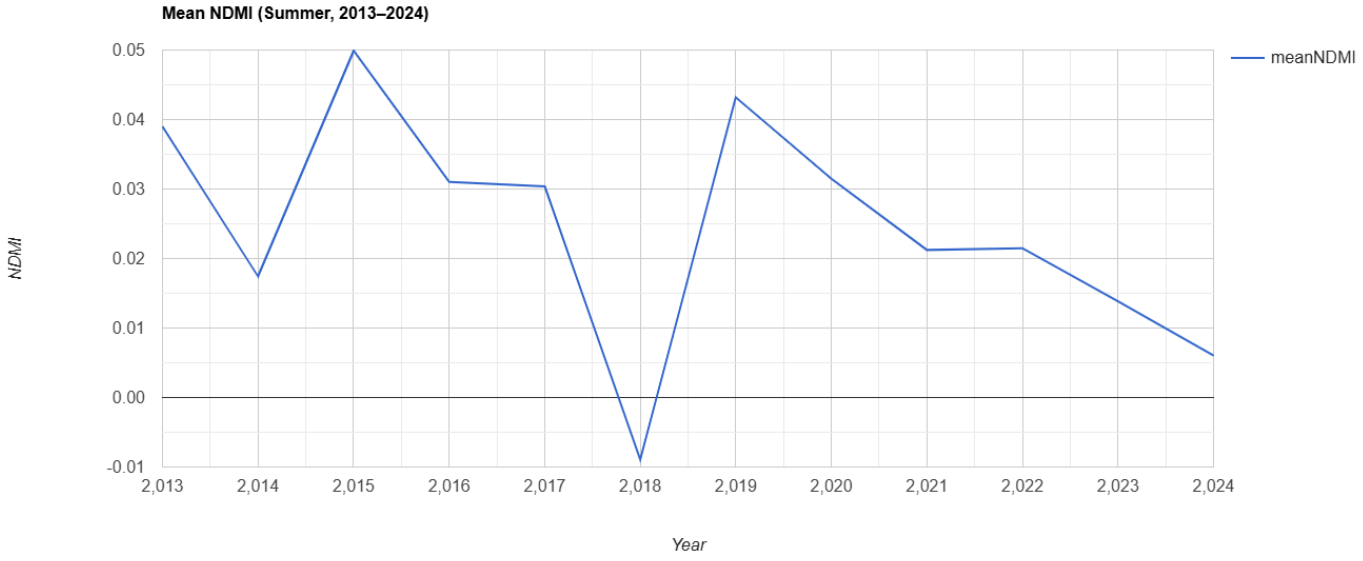

Figure 4.3b Mean Summer NDMI 2013-2024

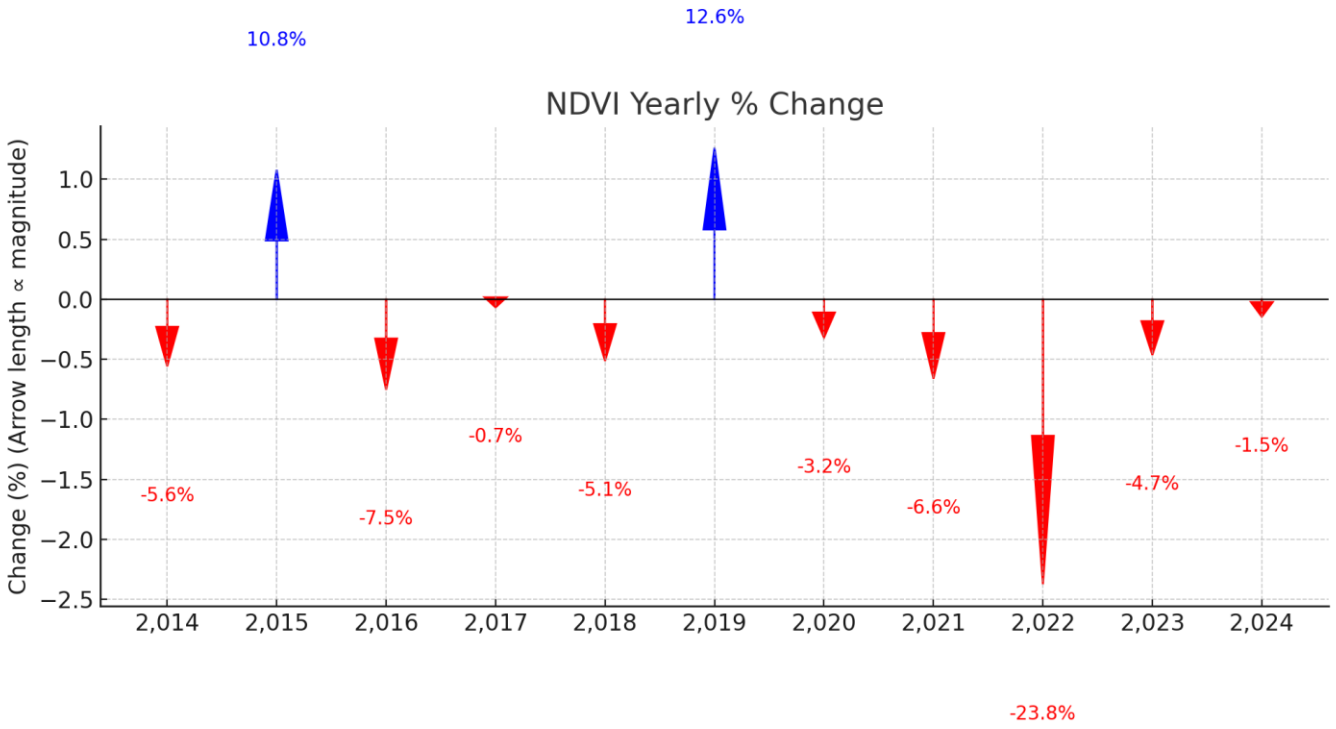

Figure 4.3c Yearly percent change in NDVI

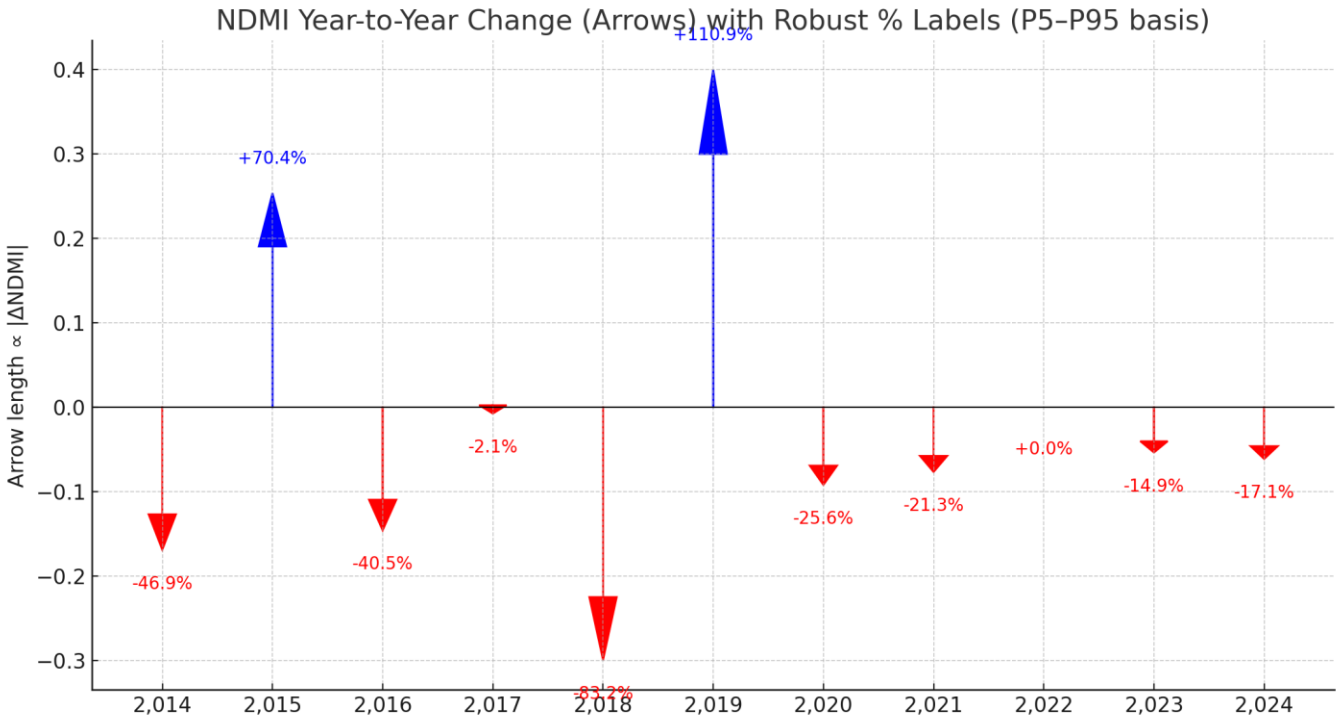

Figure 4.3d Yearly percent change in NDMI

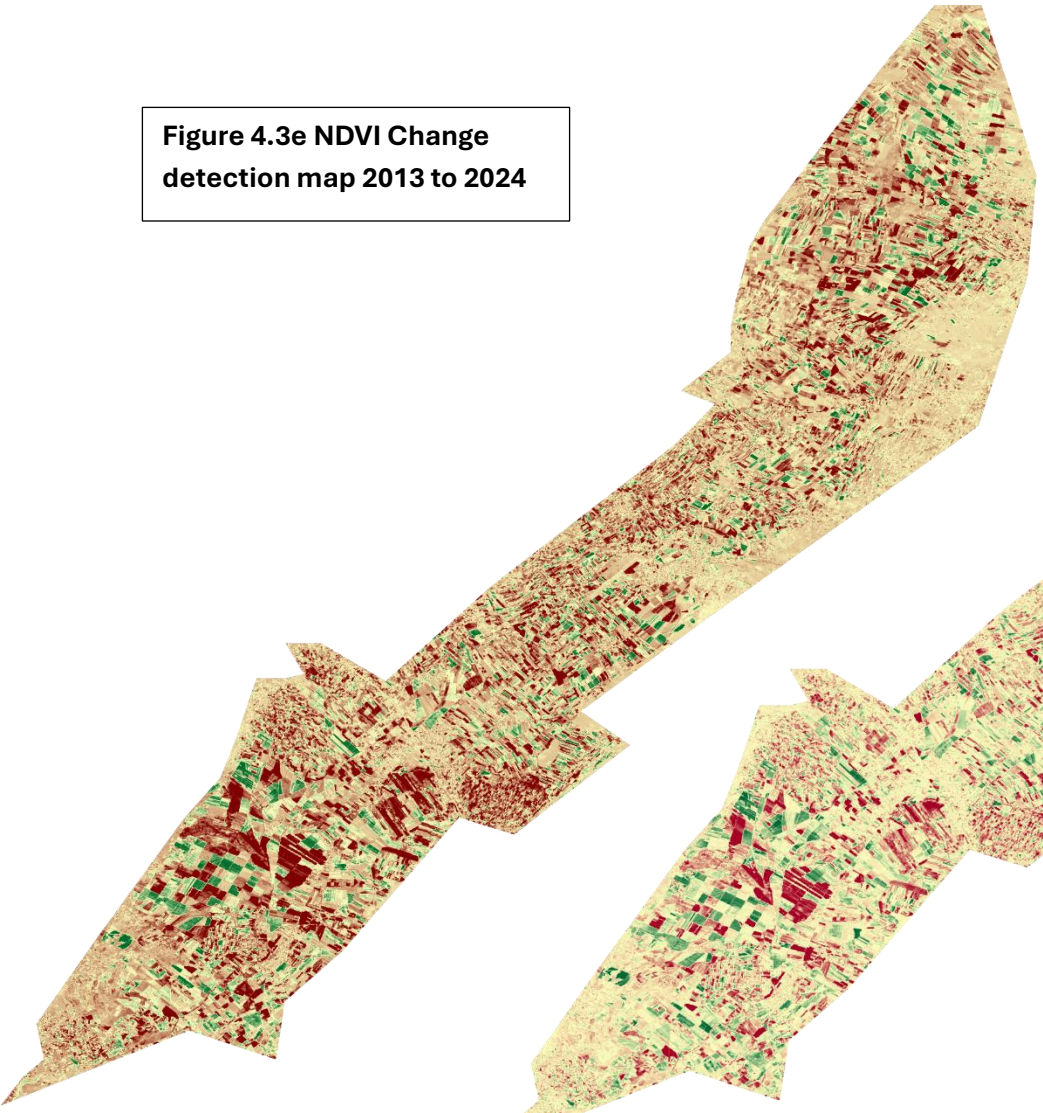

Figure 4.3e NDVI Change detection map 2013 to 2024

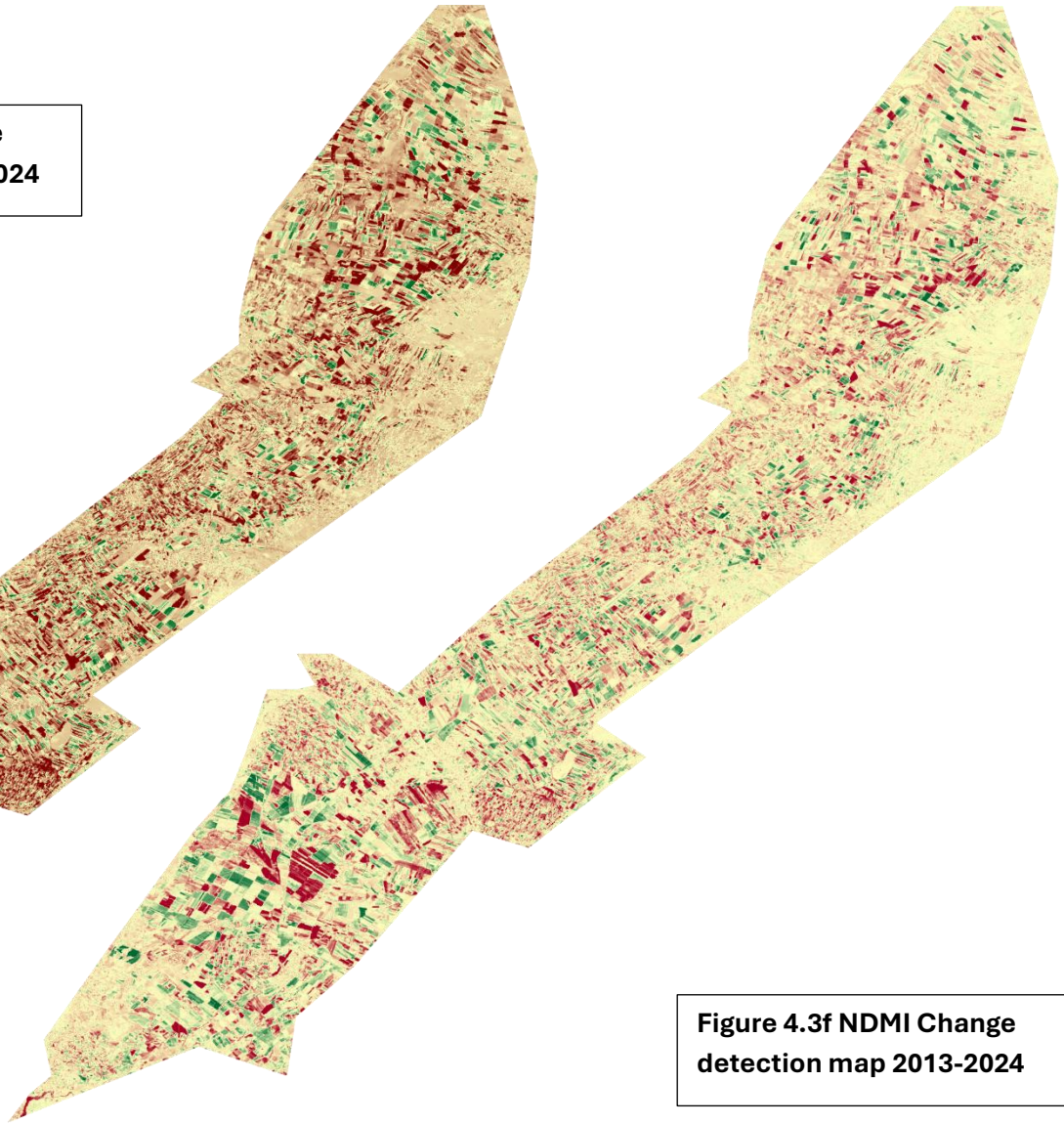

Figure 4.3f NDMI Change detection map 2013-2024

View change detection layers in GEE here

4.4 Correlations between settlement expansion and ecological stress indicators

Based on the scatter plots (figure 4.4a and 4.4b), there doesn't seem to be a clear connection between settlement area and NDVI/NDMI changes. The first major spike in hectares from 2014-2015 saw a rise in both NDVI and NDMI. The next big spike in hectares from 2016-2017 saw little to no change in NDVI/NDMI values.

The limitation of this analysis is that it only includes hectare data for 6 camps, but NDVI and NDMI values for the entire Upper Litani Basin. Using hectare data for all camps would be ideal and might show a stronger correlation between settlement area and NDMI/NDVI changes. Another limitation of this analysis is the temporal aspect of how refugee population influx corresponds to resource pressure. It might be the case that population spikes don't exert resource pressure right away, but only when that population has been sustained by the water resources for some time.

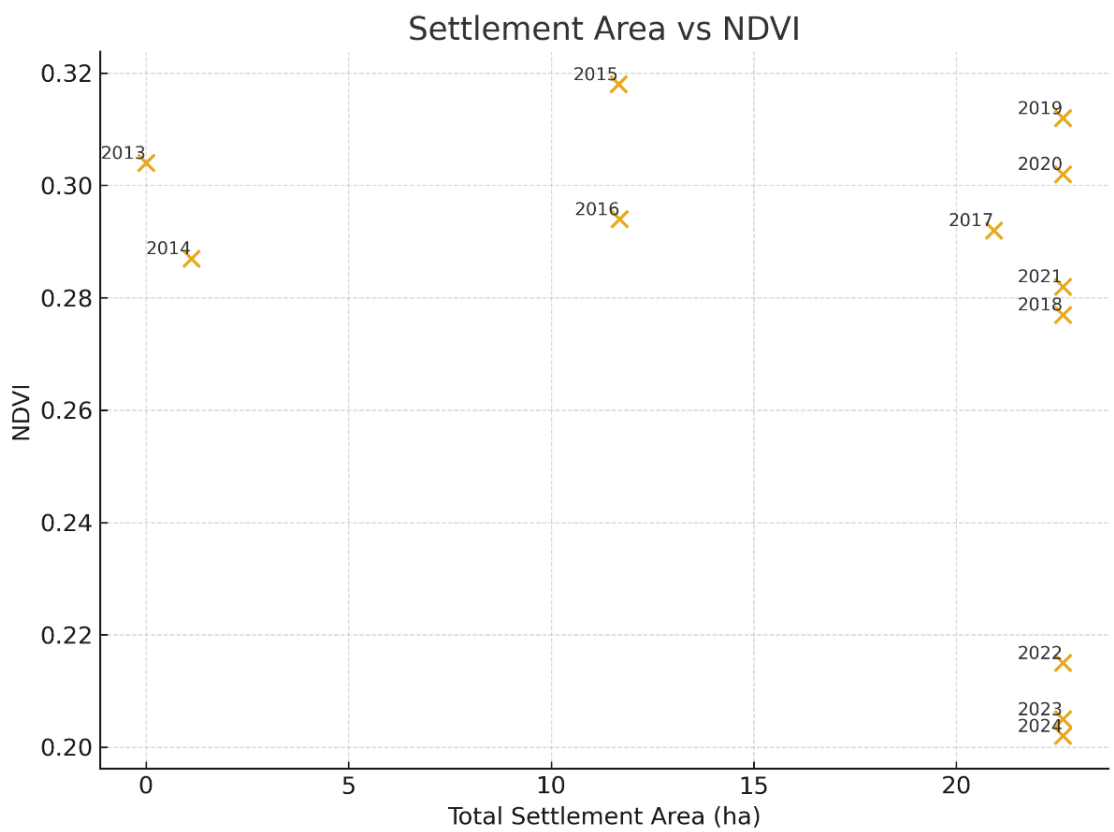

Figure 4.4a Total settlement area and NDVI

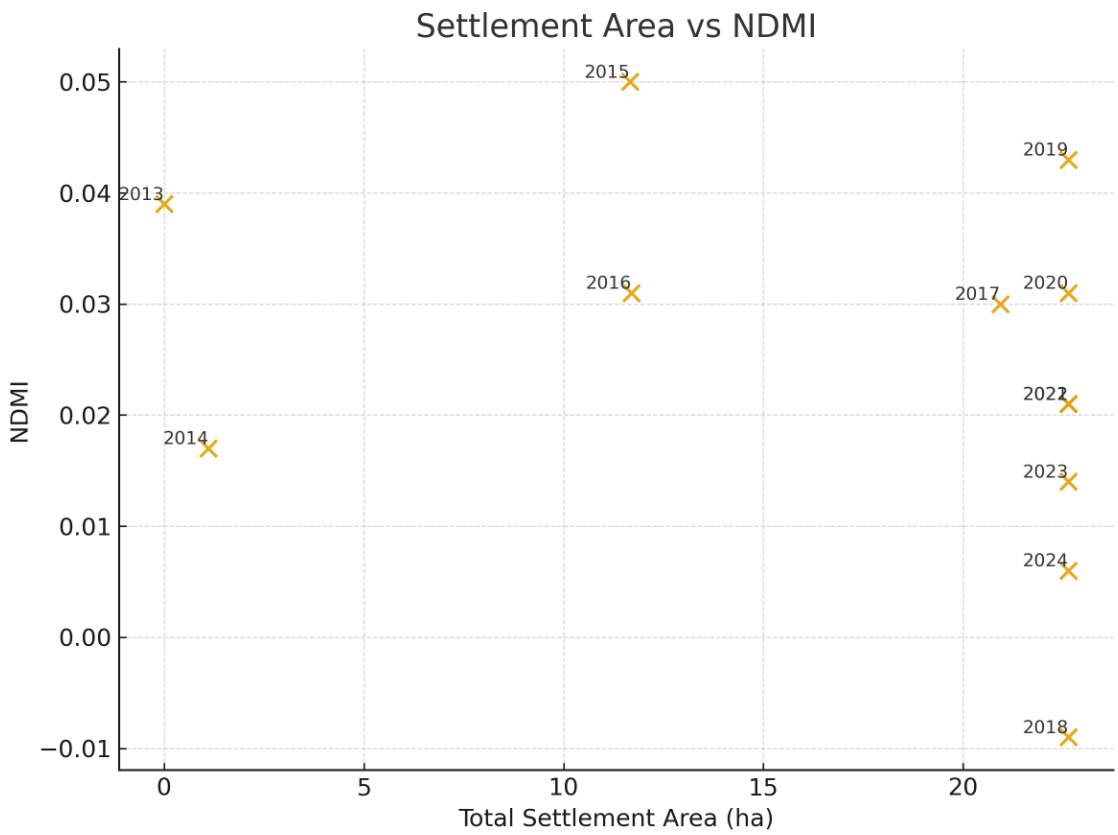

Figure 4.4a Total settlement area and NDMI

5. Conclusions

This study mapped refugee settlement growth and related environmental change in Lebanon's Bekaa Valley from 2013–2024. All six camps analyzed were built on farmland, resulting in measurable agricultural loss within 200 m buffers. NDVI and NDMI trends across the Upper Litani Basin showed stable values until 2019, followed by a sharp and sustained decline, indicating increased vegetation and water stress. The brief rise in 2018–2019 aligns with unusually high rainfall and flooding that temporarily boosted vegetation. While settlement area stabilized after 2018, the sustained decline in vegetation health suggests impacts from ongoing water demand and land-use pressure. Although no strong correlation was found between camp areas and NDVI/NDMI values, the results highlight complex environmental effects of displacement. Overall, this analysis demonstrates the usefulness of geospatial methods for assessing refugee-driven land and resource stress in fragile agricultural systems, emphasizing the need for sustained monitoring and integrated management.

References

Varouchakis, E. A., Bairaktari, K., Kampouri, K., Tsihrintzis, V. A., & Nikolakopoulos, K. G. (2025). *Spatiotemporal analysis of vegetation health and moisture using NDVI, NDMI, and SOM for monitoring of rehabilitated mining quarries.* Environmental Systems Research, 14(1), 81. https://doi.org/10.1007/s40710-025-00781-3

United Nations Economic and Social Commission for Western Asia. (n.d.). *Irrigation command area; command area for irrigation*. (Source: FAO Term Portal). Retrieved October 24, 2025, from https://www.unescwa.org/sd-glossary/irrigation-command-area-command-area-irrigation

Jaafar, H., King-Okumu, C., Haj-Hassan, M., Abdallah, C., El-Korek, N., & Ahmad, F. (2016). *Water resources within the Upper Orontes and Litani Basins: A balance, demand and supply analysis amid the Syrian refugees crisis* (IIED Working Paper). International Institute for Environment and Development. https://pubs.iied.org/10174IIED

BucknellGIS. (2023, March 8). *Lebanon informal settlements for refugees* [Feature layer]. ArcGIS REST Services. https://services2.arcgis.com/xsh7pVZv42relbEf/arcgis/rest/services/Lebanon_Informal_Settlements_for_Refugees/FeatureServer/0

Safi, A. R., Karimi, P., Mul, M., Chukalla, A., & de Fraiture, C. (2022). *Translating open-source remote sensing data to crop water productivity improvement actions*. Agricultural Water Management*, 261*, 107373. **https://doi.org/10.1016/j.agwat.2021.107373**

U.S. Geological Survey. (n.d.). *Landsat Normalized Difference Vegetation Index (NDVI)*. https://www.usgs.gov/landsat-missions/landsat-normalized-difference-vegetation-index

Jaafar, H., Archer, D., Jomaa, I., Machayekhi, D., Mansour, E., Machlab, H., & King-Okumu, C. (2016). *Balancing water stress, human crises and innovation under a changing dryland climate* (IIED Report 2016/01). International Institute for Environment and Development. http://pubs.iied.org/pdfs/17413IIED.pdf

Our World in Data. (2024). *Global precipitation anomaly* [Map dataset]. Data adapted from Copernicus Climate Change Service (ERA5 monthly averaged data on single levels). Retrieved October 24, 2025, from https://ourworldindata.org/grapher/global-precipitation-anomaly

United Nations High Commissioner for Refugees (UNHCR). (2019, January 11). *Storm flooding brings misery to Syrian refugees in Lebanon*. ReliefWeb. https://www.unhcr.org/us/news/stories/storm-flooding-brings-misery-syrian-refugees-lebanon